POWER MARKETS

Volume I Q&A Overview: Physical System, Institutions, And Market Time

Electricity markets start with a delivery problem: energy has to arrive at the right time and location through a constrained network, and deviations have to be assigned to someone.

What machine are we building?

QUESTION What has to be visible before prices, contracts, storage, and trading make sense?

ANSWER Power-market reasoning starts by connecting three layers: physical delivery, institutional responsibility, and timed market correction.

DEEPEN IN Vol I Ch 1, Ch 2, Ch 5, Ch 6, Ch 15.

1

Physical delivery

- MW is capability or demand at an instant.
- MWh is delivered energy over time.
- Frequency, voltage, congestion, and reliability turn physics into operating limits.

2

Institutional responsibility

- TSOs, DSOs, BRPs, BSPs, exchanges, suppliers, offtakers, and regulators divide the work.
- The same physical event can create different obligations for different actors.

3

Market correction

- Day-ahead, intraday, balancing, metering, and settlement are successive attempts to align expectation with delivery.
- Prices and penalties reflect time, location, uncertainty, and accountability.

When is an MWh commercially meaningful?

QUESTION What has to be specified before "one MWh" becomes a tradable or contractable thing?

ANSWER An MWh is commercially useful only when the delivery conditions are specified: time, place, metering, responsibility, and fallback if reality differs from plan.

RELATION MWh = MW × hours.

DEEPEN IN Vol I Ch 1, Ch 3, Ch 4.

1

Quantity

- Energy accumulates through time: MW multiplied by hours.
- Annual generation is a useful summary, but hourly shape determines stress and value.

2

Time

- A flat buyer, solar plant, and battery can match annually while mismatching in critical hours.
- Peaks, ramps, residual-load hours, and gate-closure times decide the relevant exposure.

3

Place

- A bidding zone, connection point, local network, loss factor, and tariff regime can change the value of the same MWh.
- Location enters through physics and through rules.

4

Responsibility

- The contract and market role decide who owns imbalance, curtailment, nomination, and settlement.
- A project can sell the same physical energy under very different risk allocations.

How does a forecast become cash?

QUESTION What happens between a forecast and the final financial result?

ANSWER The system moves from expectation to schedule to physical deviation to balancing action to settlement. Each step narrows what can still be changed.

ANCHOR A 1 percent miss on a 500 MW weather-dependent portfolio is a 5 MW operating deviation in that hour.

DEEPEN IN Vol I Ch 2, Ch 6, Ch 8.

1 Forecast

- Weather, demand, outage, and price expectations estimate the next delivery period.
- Forecast quality matters only through decisions it changes.

2 Schedule

- Expected production and consumption become market positions or nominations.
- The schedule creates a baseline for later comparison.

3 Deviation

- Actual generation and demand diverge from the plan.
- The deviation can be physical, financial, contractual, or all three.

4 Balancing

- The TSO uses available resources to keep the system secure in real time.
- Balancing actions occur after many commercial choices have already expired.

5 Settlement

- Metered reality is converted into money and accountability.
- The relevant question becomes who owned the deviation and what information was available.

Why can a strong resource be a weak project?

QUESTION Why can a sunny or windy site lose to a less impressive site?

ANSWER Resource quality creates potential MWh. Grid access decides whether those MWh can be injected, priced, and monetized without excessive delay or curtailment.

RELATION Grid exposure has MW terms, MWh terms, and curtailment terms; each scales differently.

DEEPEN IN Vol I Ch 4; Vol II Ch 2; Case C03.

1

Resource

- Irradiation, wind speed, availability, and degradation set the production profile.
- Nameplate capacity alone says little about delivered value.

2

Connection right

- Export and import capacity define what the asset can physically do at the node.
- Queue timing and reinforcement risk can dominate yield.

3

Network constraint

- Thermal, voltage, stability, and congestion constraints can limit output or flexibility.
- The relevant bottleneck can be local even when the market price is zonal.

4

Tariff and loss regime

- Capacity charges, MWh charges, losses, and curtailment compensation alter dispatch economics.
- A lower-yield site with cleaner grid rights can rank higher.

Who owns each failure mode?

QUESTION Which actor is responsible when the system misses plan, needs flexibility, or settles a deviation?

ANSWER The actor map is a map of accountability. It shows where technical failure becomes financial exposure, operational duty, or rule enforcement.

DEEPEN IN Vol I Ch 5, Ch 8, Ch 9; Vol II Ch 15.

System operators

- TSO: system security, balancing, reserve procurement, grid-code requirements.
- DSO: local network operation, connection, queue, local tariff and export/import limits.

Commercial responsibility

- BRP: imbalance responsibility, schedules, settlement exposure.
- BSP: qualified balancing-service provision, reserve capacity, activation delivery.

Market venues

- NEMO / exchange: day-ahead and intraday market access and clearing.
- Market operator rules define what can be traded and when.

Contract parties

- Supplier / offtaker: procurement shape, settlement basis, customer promise, residual exposure.
- Asset owner: production, data, dispatch rights, and retained market exposure.

Rule layer

- Regulator: market conduct, allowed revenues, tariff framework, consumer/system protections.
- Compliance and surveillance shape what strategies remain acceptable.

Project implication

- A project model needs a role map before it can assign forecast error, curtailment, reserve delivery, credit, and settlement risk.

Why does the market have a clock?

QUESTION Why are there several markets for the same delivery hour?

ANSWER Electricity is traded before delivery but balanced during delivery and settled after delivery. The clock matters because information, liquidity, and rights change over time.

ANCHOR Moving from 60-minute to 15-minute market time units turns one delivery hour into four settlement/trading intervals.

DEEPEN IN Vol I Ch 6, Ch 8, Ch 9; Vol III Ch 2.

1

Before day-ahead

- Development, hedging, PPA negotiation, and maintenance planning set the long-horizon position.
- Many important choices are already embedded before the operating day.

2

Day-ahead

- The next delivery day receives a common schedule and price reference.
- It is the main coordination step for portfolio planning and contract references.

3

Intraday

- Forecast updates become tradable if liquidity, access, and gate closure allow adjustment.
- Better information has value only if action is still possible.

4

Balancing

- The TSO activates resources for residual system needs close to real time.
- Reserve qualification and activation rules decide which assets can participate.

5

Settlement

- Metering and settlement convert delivery into invoices, imbalance costs, and performance evidence.
- Partner performance can be judged only against feasible actions at the relevant time.

Why can cheap solar earn a weak price?

QUESTION If solar has low variable cost, why can more solar reduce solar revenue in sunny hours?

ANSWER Low variable cost helps solar clear the market. Realized revenue still depends on the price in the hours when solar produces, and correlated output can push those prices down.

RELATION Capture price = sum(price_t × generation_t) / sum(generation_t).

DEEPEN IN Vol I Ch 7, Ch 12; Vol II Ch 6.

Merit-order logic

- Offers are stacked until demand is met.
- The marginal accepted resource sets the simplified clearing price.
- A low-cost plant can receive a price set by a more expensive resource when it produces in scarce hours.

Shape and cannibalization

- Solar production is synchronized by daylight and weather.
- When similar output arrives together, the marginal unit can become cheaper and sunny-hour prices can fall.
- Curtailment, negative prices, and capture-price erosion come from shape, location, and flexibility.

Why do curves beat annual totals?

QUESTION What information disappears when a project is summarized as annual MWh?

ANSWER Annual energy hides the hours that drive capacity, congestion, prices, balancing cost, and contract exposure.

RELATION Residual load = demand - variable renewable output, using the chosen resource convention.

DEEPEN IN Vol I Ch 3, Ch 12; Vol II Ch 6.

Load curve

- Peak demand drives capacity need, grid stress, and scarcity.
- Average demand can hide the hours that require the system to work hardest.

Residual load

- Demand minus variable renewable output shows what flexible resources still need to cover.
- The residual curve reveals where batteries, hydro, demand response, and thermal units become valuable.

Contract shape

- Buyers can ask for flat, shaped, as-consumed, or hourly matched supply.
- Each higher promise assigns mismatch risk somewhere.

Project model

- The useful model preserves time resolution long enough to see scarcity, curtailment, and settlement exposure.
- The core reasoning step preserves hourly exposure; annual totals summarize afterward.

What does the grid buy besides energy?

QUESTION Why are system services separate from ordinary MWh delivery?

ANSWER Reliability needs both delivered energy and available capability. Some products pay for being ready; others pay for instructed movement or stability behavior.

ANCHOR Reserve capacity is a MW availability product; balancing energy is instructed MWh movement.

DEEPEN IN Vol I Ch 9, Ch 10; Vol III Ch 8.

Product map

Product family	Unit that matters	System need	Project question
Energy	MWh	Serve demand over an interval	What price applies in production hours?
Reserve capacity	MW held available	Keep response ready before activation	What headroom, sustain, and qualification are required?
Balancing energy	MWh activated	Correct residual imbalance	What action was instructed and delivered?
Stability services	capability / MW / technical tests	Frequency, voltage, fault behavior, inertia-like response	Can the asset qualify and be measured?

How do technologies carry different exposure?

QUESTION Why do two assets with the same annual MWh face different market problems?

ANSWER Each technology has a physical pattern: controllability, correlation, fuel or resource dependence, duration, and grid interaction. Those patterns become commercial exposure.

DEEPEN IN Vol I Ch 11, Ch 12, Ch 13; Vol III Ch 13.

Technology map

Technology	Physical pattern	Main exposure
Solar	Daylight-correlated, weather-dependent	Capture-price pressure, curtailment, shaping need
Wind	Weather-dependent, regional correlation	Forecast error, correlated output, balancing exposure
Hydro	Stored energy where reservoirs exist	Intertemporal optimization and flexibility value
Thermal	Dispatchable, fuel-linked	Marginal pricing, scarcity value, fuel and emissions exposure
Battery	Finite power and energy, fast response	Time-shifting, reserves, SoC, degradation, grid rights
Demand response	Flexible consumption with operational limits	Local flexibility, adequacy support, customer constraints

What is a battery in market terms?

QUESTION Why is a battery more than an energy bucket?

ANSWER A battery is a bundle of scarce resources: MW, MWh, state of charge, cycles, connection rights, control rights, and safety/warranty constraints.

RELATION Duration in hours = usable MWh / MW; round-trip efficiency turns gross spread into smaller net margin.

DEEPEN IN Vol I Ch 13; Vol II Ch 8, Ch 9, Ch 12; Cases C02, C04.

Power

- MW rating governs charge, discharge, reserve commitment, and connection use.
- Power is the binding unit for many capacity-style services.

Energy

- MWh capacity determines duration and stress-period coverage.
- Extra duration matters only when the marginal hours are valuable enough.

State of charge

- Today's action changes tomorrow's optionality.
- Reserve promises and energy arbitrage compete for the same headroom.

Asset life

- Round-trip efficiency, degradation, warranties, insurance, and safety rules convert gross spread into margin.

Rights

- Grid import/export limits and dispatch rights decide which theoretical values can actually be captured.

How do Nordic examples translate into project decisions?

QUESTION What makes the Nordic region useful for learning without turning regional averages into project answers?

ANSWER The Nordic system is integrated enough to show hydro, interconnection, weather, bidding zones, and balancing reform interacting; project decisions remain country, zone, connection, and product specific.

DEEPEN IN Vol I Ch 4, Ch 5, Ch 6, Ch 9, Ch 14.

Regional mechanism

- Hydro, interconnectors, wind, nuclear, thermal capacity, and growing solar interact differently by zone.
- Weather correlation and transmission constraints can dominate short-term prices.
- Bidding zones make network constraints visible at a coarse level.

Project translation

- Tariffs, connection practice, BRP/BSP implementation, reserve eligibility, and local network limits remain jurisdictional.
- A Nordic opportunity often starts as a grid-right and product-eligibility question before it becomes a market-price question.
- Regional statistics orient the analysis; site economics come from the local connection and role structure.

Which checks survive into commercial work?

QUESTION Before accepting a market claim, what has to be pinned down?

ANSWER A useful power-market claim names the hour, place, binding constraint, responsible actor, market interval, and contract owner.

DEEPEN IN Vol I Ch 15; Vol II Ch 14, Ch 15; Vol III Ch 12.

1. **Hour and location**
   - Which delivery hour, bidding zone, and connection point does the claim describe?
2. **Binding constraint**
   - Is the limiting object MW, MWh, ramp, voltage, frequency, network capacity, data, or credit?
3. **Responsible actor**
   - Which party owns the deviation, settlement, service delivery, or customer promise?
4. **Market interval**
   - Could day-ahead, intraday, balancing, or settlement still change the position?
5. **Contract owner**
   - Which contract clause assigns shortfall, surplus, curtailment, fallback, certificate evidence, or control rights?